

	YTU – Faculty of Chemical and Metallurgical Engineering, Mathematical Engineering Department				NOTE CHART										
					1. q		2. q		3. q		Total				
1st MIDTERM EXAM															
Student Name and Surname															
Students Number				Signature				Exam Date		16/11/2017					
Course Name		Linear Algebra		Group Number		2		Exam Duration				Examination Room			
Course Instructor Name and Surname				Assoc. Prof. Dr. FATMA AYDIN AKGÜN											
Student Disciplinary Regulations "and to make or attempt to make copies of exams to" the actual perpetrators are suspended from one or two semesters. (YÖK; 2547 Student Disciplinary Regulations, 9. Article)															

1) Prove the following statements.(30p)

a) If $A^T A = A$ then A is symmetric and idempotent.

b) If A is orthogonal matrix, then A^2 is also orthogonal and $((\det(A))^2 = 1$.

(15) a) $A = A^T$ (5)
 $A^T = (A^T A)^T = A^T A = A$ Thus $A^T = A \rightarrow A$ is symmetric. (5)
 $A^2 = A \cdot A = A^T \cdot A = A$ $A^2 = A$ A is idempotent. (5)

b) A is orthogonal
then $A^{-1} = A^T$ (3)

? A^2 is orthogonal $(A^2)^{-1} = (A^2)^T$ (7)
 $A^2 = A \cdot A$ $(A^2)^{-1} = (A \cdot A)^{-1} = \underbrace{A^{-1}}_{A^T} \cdot \underbrace{A^{-1}}_{A^T} = (A A^T)^T = (A^2)^T$

2)(40p) $V \subset \mathbb{R}^4$, $V = \{(a, b, c, d) : a = d, b = 2c\}$

a) Show that V is a subspace.

b) Find the basis of V .

c) Find the dimension of V .

A orthogonal ise $A^T A = I$
 $|A^T A| = |I| = 1$ (5)
 $|A|^2 = 1$ //

c) Find the coordinates of the vector $u = \begin{bmatrix} 3 \\ 4 \\ 2 \\ 3 \end{bmatrix}$ according to the basis of V obtained in b). a-d=0 b-2c=0

a) To be subspace $(0,0,0,0) \in V$?

$(0,0,0,0) \in V$ then

i) $v_1 + v_2 \in V$ where $v_1 \in V$ $v_2 \in V$

ii) $kv_1 \in V$ where k is any real number.

$v_1 = \begin{bmatrix} a_1 \\ 2c_1 \\ c_1 \\ a_1 \end{bmatrix}$ $v_2 = \begin{bmatrix} a_2 \\ 2c_2 \\ c_2 \\ a_2 \end{bmatrix}$

$v_1 + v_2 = \begin{bmatrix} a_1 + a_2 \\ 2(c_1 + c_2) \\ c_1 + c_2 \\ a_1 + a_2 \end{bmatrix}$

where $v \in V$ so

$kv_1 = \begin{bmatrix} ka_1 \\ 2kc_1 \\ kc_1 \\ ka_1 \end{bmatrix}$

$v_{11} = v_{14}$ $v_{12} = 2v_{13}$ $v \in V$

the basis are $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix}$

dim $V = 2$.
 $\alpha_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 2 \\ 3 \end{bmatrix}$ $\alpha_1 = 3$ $\alpha_2 = 2$.
 $3b_1 + 2b_2 = u$ where b_1, b_2 basis.

1)
$$A = \begin{bmatrix} a_1 + b_1 t & a_2 + b_2 t & a_3 + b_3 t \\ a_1 t + b_1 & a_2 t + b_2 & a_3 t + b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} = ? \quad (1-t^2) \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

denklemin kökleri bilinir

Ters yok ise t 'yi bulunuz.

2)
$$\begin{matrix} x_1 & 3x_1 + x_3 & y \\ 2x_1 - x_2 & & \end{matrix} \rightarrow y$$

$$\begin{cases} 4x + 5y = 2 \\ 11x + y + 2z = 3 \\ x + 5y + 2z = 1 \end{cases}$$

bu denklemleri kullanarak
denklemin tersi yöntemini veya
Cramer yöntemi kullanarak
çözünüz.

$$x = \frac{3}{11}, \quad y = \frac{2}{11}, \quad z = -\frac{1}{11}$$

3) $\mathbb{R}^3$ uzayında u - vektörüne sahip olan

$$\langle u, v \rangle = u_1 v_1 + 2u_2 v_2 + 3u_3 v_3$$

Gru - Schmidt kullanarak $u_1(1,1,1)$, $u_2(1,1,0)$
 $u_3(1,0,0)$ bir ortonormal
bazı oluştur.

Carp
$$\left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right) \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}} \right) \left(\frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, 0 \right)$$

Örnek: $L: P_2 \rightarrow P_3$

$$L(p(x)) = x^2 \frac{d}{dx} p(x) = x^2 p'(x)$$

şeklinde tanımlanan lineer dönüşümün) çekirdeğini, çekirdeğinin
bir bazını ve $\text{boy Ker } L = ?$ bulunuz.

b) $L(P_2)'$ nin bir bazını ve $\text{boy } L(P_2) = ?$ bulunuz.

Çözüm: a) $p(x) = ax^2 + bx + c$, $L(p(x)) = 0_{P_3}$ den

$$L(ax^2 + bx + c) = 0 \Rightarrow ax^2 + bx + c \in \text{Ker } L \text{ dir.}$$

$$x^2 p'(x) = x^2 (2ax + b) = 0$$

$$\Downarrow \\ a=0, b=0, c \in \mathbb{R}$$

$$\text{Ker } L = \{ (0, 0, c) \}, c=1 \text{ alınırsa}$$

$$\text{Ker } L = \{ (0, 0, 1) \} \Rightarrow \text{boy Ker } L = 1 \text{ dir.}$$

$\text{Ker } L$ 'nin bir bazı $\{ (0, 0, 1) \}$ dir.

b) ~~is it a basis or not~~

$$\begin{aligned} \text{c) } L(P_2) &= x^2 (ax^2 + bx + c)' = x^2 (2ax + b) \\ &= 2ax^3 + bx^2 \end{aligned}$$

olduğundan $\{ x^2, x^3 \}$, $L(P_2)$ 'yi gerer. Bu küme lineer bağımsız
olduğundan aynı zamanda bir bazdır.

$$\text{Boy } L(P_2) = 2 \text{ dir.}$$

$$\begin{aligned} \text{c) } \underbrace{\text{Boy Ker } L}_1 + \underbrace{\text{Boy } L(P_2)}_2 &= \text{Boy}(P_2) \\ &= 3 \end{aligned}$$

P_2 nin doğal bazı $\{ 1, x, x^2 \}$ dir.

is it a basis or not

3) $u_1 = v_1$

$$v_2 = u_2 - \frac{(u_2, v_1)}{(v_1, v_1)} v_1$$

$$\langle u_{2,5} \rangle = 1 + 2 = 3$$

$$= \left(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2} \right) \quad \checkmark$$

$$\langle \sigma_1, \sigma_1 \rangle = 6$$

$$v_3 = u_3 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2 = (1, 0, 0) - \frac{1}{6} (1, 1, 1) - \frac{1}{3} \left(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2} \right)$$

$(u_{3,5}) = 1$

$$\langle u_{3,52} \rangle = \frac{1}{2}$$

$$\langle \sigma_{2, \sigma_2} \rangle = \frac{1}{4} + 2 \frac{1}{4} + 3 \frac{1}{4} = \frac{6}{4}$$

$$w_1 = \frac{v_1}{\|v_1\|} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) \quad (3) \quad \checkmark$$

$$\omega_2 = \frac{v_2}{11.5211} = \left(\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right)$$

$$\omega_3 = \left(\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}, 0 \right)$$

$$= \left(\frac{4}{6}, -\frac{2}{6} \right)$$

$$\langle v_3, v_3 \rangle = \|v_3\|^2 = \frac{4}{9} + 2\frac{1}{9} = \frac{6}{9}$$

$$\begin{array}{r} 16 \\ \hline 36 \end{array} \frac{14}{36} \quad \begin{array}{r} 16 \\ \hline 36 \end{array} \frac{14}{36} \quad \begin{array}{r} 16 \\ \hline 36 \end{array} \frac{14}{36} \quad \begin{array}{r} 16 \\ \hline 36 \end{array} \frac{14}{36}$$

a) $\lim_{x \rightarrow 0} \frac{x}{\sin(\sin x)} =$	
b) $\lim_{x \rightarrow \frac{\pi}{2}} e^{\tan x} =$	d) $\lim_{x \rightarrow 0} \frac{\sqrt{2x^2 + 4x + 1} - \sqrt{x^2 - x + 1}}{2x}$
c) $\lim_{x \rightarrow \infty} \sqrt{x^2 + x + 1} - \sqrt{x^2 - x + 1} =$	e) $\lim_{n \rightarrow -\infty} \frac{7^n + 3^n + 5}{3n - 7n + 25}$

3- Compute the following limits (35p)

2)

$$4x + 5y = 2$$

$$11x + y + 2z = 3$$

$$x + 5y + 2z = 1$$

} Lin. denk. sistem.
matrisin tersi veya Cramer metoduyla
çözülür.

$$|A| = \begin{vmatrix} 4 & 5 & 0 \\ 11 & 1 & 2 \\ 1 & 5 & 2 \end{vmatrix} = 2 \begin{vmatrix} 4 & 5 & 0 \\ 11 & 1 & 1 \\ 1 & 5 & 1 \end{vmatrix} \quad (5)$$

$$= 2 \begin{vmatrix} 4 & 5 & 0 \\ 10 & -4 & 0 \\ 1 & 5 & 1 \end{vmatrix} = 2(-16 - 50) = -132$$

$$A_1 = \begin{vmatrix} 2 & 5 & 0 \\ 3 & 1 & 2 \\ 1 & 5 & 2 \end{vmatrix} = 2 \begin{vmatrix} 2 & 5 & 0 \\ 3 & 1 & 1 \\ 1 & 5 & 1 \end{vmatrix} \quad (5)$$

$$= 2 \begin{vmatrix} 2 & 5 & 0 \\ 2 & -4 & 0 \\ 1 & 5 & 1 \end{vmatrix} = 2(-8 - 10) = -36$$

$$A_2 = \begin{vmatrix} 4 & 2 & 0 \\ 11 & 3 & 2 \\ 1 & 1 & 2 \end{vmatrix} = 2 \begin{vmatrix} 4 & 2 \\ 11 & 3 \\ 1 & 1 \end{vmatrix} \quad (5)$$

$$= \begin{vmatrix} 0 & 2 & 0 \\ 5 & 3 & 2 \\ -1 & 1 & 2 \end{vmatrix} = -2 \begin{vmatrix} 5 & 2 \\ -1 & 2 \end{vmatrix} = -24$$

$$A_3 = \begin{vmatrix} 4 & 5 & 2 \\ 11 & 1 & 3 \\ 1 & 5 & 1 \end{vmatrix} = \begin{vmatrix} 2 & -5 & 2 \\ 8 & -14 & 3 \\ 0 & 0 & 1 \end{vmatrix} \quad (5)$$

$$= -28 + 10 = -18$$

$$x_1 = \frac{3}{11}$$

$$x_2 = \frac{2}{11}$$

$$x_3 = \frac{-1}{11} \quad (25)$$

$$A^c = \begin{bmatrix} -8 & 20 & 54 \\ -10 & 8 & -15 \\ 10 & -8 & -51 \end{bmatrix} \quad (10)$$

$$A^* = \begin{bmatrix} -8 & -10 & 10 \\ -20 & 8 & -8 \\ 54 & -15 & -51 \end{bmatrix} \quad (5)$$

$$A^{-1} = \frac{A^*}{|A|} = \begin{bmatrix} -8 & -10 & 10 \\ -20 & 8 & -8 \\ 54 & -15 & -51 \end{bmatrix} \cdot \frac{1}{132} \quad (5)$$

$$X = \frac{1}{132} \begin{bmatrix} -8 & -10 & 10 \\ -20 & 8 & -8 \\ 54 & -15 & -51 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \quad (5)$$

$$= \begin{bmatrix} \frac{3}{11} \\ \frac{2}{11} \\ \frac{-1}{11} \end{bmatrix} \quad (5)$$

$$-16 - 30 + 10$$

$$1) A = \begin{bmatrix} a_1 + b_1 t & a_2 + b_2 t & a_3 + b_3 t \\ a_1 t + b_1 & a_2 t + b_2 & a_3 t + b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \quad \text{var}$$

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~~Tersiyi yok ise~~

~~t'yi bulamaz~~
kalkarsa buluruz -

Cevap

A'nın tersiyi yok ise $|A| = 0$.

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$$\begin{vmatrix} a_1 + b_1 t & a_2 + b_2 t & a_3 + b_3 t \\ a_1 t + b_1 & a_2 t + b_2 & a_3 t + b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ a_1 t + b_1 & a_2 t + b_2 & a_3 t + b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} - t \begin{vmatrix} a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$+ \begin{vmatrix} b_1 t & b_2 t & b_3 t \\ a_1 t + b_1 & a_2 t + b_2 & a_3 t + b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

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$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + t^2 \begin{vmatrix} b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$(1 - t^2) \underbrace{\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}}_{|A'|} = 0.$$

$$t = \pm 1$$

$$|A| = 0$$

//

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